

Real and Complex Analysis General Exam Fall 2007

Problem 1

Suppose that $(X, \mathcal{A}, \mu)$ is a measure space, with $\mathcal{A}$ a σ -algebra, μ a finite measure. Let $\mathcal{A}_0 \subset \mathcal{A}$ be a sub-algebra of sets which generates $\mathcal{A}$; thus $\mathcal{A}$ is the smallest σ -algebra generated by $\mathcal{A}_0$.

Let $\mathcal{B} \subset \mathcal{A}$ be the collection of subsets of X with the property that for every ϵ and $B \in \mathcal{B}$, there is an $A \in \mathcal{A}_0$, with

$$\mu(A \Delta B) < \epsilon.$$

(Here, Δ is the symmetric difference, $C \Delta D = (C - D) \cup (D - C)$.)

Show that $\mathcal{B}$ is a σ -algebra containing $\mathcal{A}_0$, in particular that it is an algebra, and that it is closed under countable (increasing) unions.

Hint: You may assume that (do not prove!):

$$\begin{aligned}\mu(C \Delta D) &= \mu(C^c \Delta D^c) \\ \mu((D_1 \cup D_2) \Delta (E_1 \cup E_2)) &\leq \mu(D_1 \Delta E_1) + \mu(D_2 \Delta E_2) \\ |\mu(C \Delta E) - \mu(D \Delta E)| &\leq \mu(C \Delta D)\end{aligned}$$

Problem 2

(a) Let $f(x)$ be a real-valued differentiable function on a neighborhood of $[a, b]$, and assume that

$\frac{df}{dx}(a) < 0$, and $\frac{df}{dx}(b) > 0$. Show that there exists a $c \in (a, b)$ such that $\frac{df}{dx}(c) = 0$.

(b) Suppose again that $\frac{df}{dx}(a) < \frac{df}{dx}(b)$ and that λ satisfies $\frac{df}{dx}(a) < \lambda < \frac{df}{dx}(b)$. Show that there exists a $c \in (a, b)$ with $\frac{df}{dx}(c) = \lambda$.

Problem 3

Let (X, μ) be a measure space.

(a) Show that if g is a non-negative square-integrable function on X , then

$$\int_{\{x: g(x) \geq \lambda\}} g(x) d\mu(x) \leq \frac{1}{\lambda} \int_X g^2(x) d\mu(x)$$

Suppose now that (X, μ) is a finite measure space and let $\{f_n\}_{n=1,2,\dots}$ be a sequence of non-negative functions which are both integrable and square-integrable, i.e., in $L^1(X) \cap L^2(X)$, that they converge pointwise to a function $f(x)$, and that the limits

$$\lim_{n \rightarrow \infty} \int f_n(x) d\mu(x) = L \text{ and } \lim_{n \rightarrow \infty} \int f_n^2(x) d\mu(x) = M$$

exist.

(b) Show that

$$\lim_{n \rightarrow \infty} \int f_n(x) d\mu(x) = \int f(x) d\mu(x)$$

Hints: Why is

$$\int f(x) d\mu(x) \leq \liminf \int f_n(x) d\mu(x)?$$

To show an inequality in the other direction: Define cut-off functions:

$$f_{n,\lambda}(x) \equiv \begin{cases} f_n(x), & f_n(x) \leq \lambda \\ \lambda & \text{otherwise} \end{cases}$$

which converge pointwise to

$$f_\lambda(x) \equiv \begin{cases} f(x), & f(x) \leq \lambda \\ \lambda & \text{otherwise} \end{cases}$$

Show that, given ϵ , there is a λ such that (again, μ is a finite measure and use part (a))

$$\begin{aligned} \limsup_n \int f_n(x) d\mu(x) &\leq \limsup_n \int f_{n,\lambda}(x) d\mu(x) + \epsilon \\ &\leq \int f_\lambda(x) d\mu(x) + \epsilon \end{aligned}$$

which is clearly

$$\leq \int f(x) d\mu(x) + \epsilon$$

Problem 4

Let f be an entire function on the complex plane, and suppose there is a positive integer N such that

$$\frac{f(z)}{z^N} \rightarrow 0 \text{ as } |z| \rightarrow \infty$$

Find an upper bound on the number of zeros of f (counting multiplicity) in the complex plane in terms of N . Is the bound sharp?

Problem 5

Let $\mathcal{H}$ be the Hilbert space of $L^2(D, \mu)$ functions on the unit disc $D = \{z \in \mathcal{C} : |z| < 1\}$, with μ two-dimensional measure, $d\mu = dxdy = r dr d\theta$ in polar coordinates, with $\|f\|_{L^2(D)}$ its L^2 -norm. Suppose that $f \in \mathcal{H}$ is moreover analytic in D .

(a) Show that, for $0 < r < 1$

$$f(0) = \frac{1}{2\pi} \int_0^{2\pi} f(re^{i\theta}) d\theta$$

and that for $0 < r_1 < 1$,

$$f(0) = \frac{1}{\pi r_1^2} \int_0^{2\pi} \int_0^{r_1} f(re^{i\theta}) r dr d\theta$$

(b) Show that

$$|f(0)| \leq \frac{1}{\sqrt{\pi} r_1} \|f\|_{L^2(D)}$$

for all $f \in \mathcal{H}$ which are analytic in D .

Problem 6

Use the method of residues to evaluate

$$\int_{-\infty}^{\infty} \frac{x \sin x}{x^2 + 4} dx$$

Show all estimates.

Problem 7

Suppose that f is analytic on $A = \{z \in \mathbb{C} : 1 \leq |z| \leq 3\}$, and assume that $|f(z)| \leq 1$ for $|z| = 1$ and $|f(z)| \leq 9$ for $|z| = 3$. Prove that $|f(2i)| \leq 4$.